Scalability&FuturePlan

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ProjectName	**Electric Vehicle Market Performance & Insights**
MaximumMarks	1Mark

Scalability&FuturePlan:

Thisdocumentoutlineshowthecurrentprojectsolutioncanbescaledtohandlelargeruserbases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

CurrentSystemLimitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Static Local Data Extracts: Raw EV data is imported from manual spreadsheets (.csv/.xlsx) stored locally on your desktop.	New global vehicle launches, market sales, and battery specs must be manually updated and re-extracted into the workbook.	High
2	**Desktop-Bound Workbook Hosting:** The project lives primarily inside a localized .twbx workbook format.	Reviewers must open the physical file in Tableau Desktop or Tableau Reader, restricting frictionless stakeholder collaboration.	Medium
3	**Row-Level Calculation Performance:** Heavy Level of Detail (LOD) formulas run on the client side at runtime.	As millions of row-level records are added to the dataset, dashboard rendering speeds will degrade during filter actions.	Medium

ScalabilityPlan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local file sharing or single-user view via Tableau Public.	Publish workbook to **Tableau Server** using Row-Level Security (RLS) to manage hundreds of users simultaneously.
2	Data Storage	Bound by local desktop hard drive space using static flat files.	Migrate backend data assets to a cloud database (e.g., **Snowflake)** and connect directly via Tableau.
3	Performance	Desktop RAM processing client-side dashboard queries.	Implement a **Tableau Hyper Extract Schedule** or pre-aggregate heavy calculations upstream in the database before Tableau reads it.
4	Security	Open .twbx file where anyone can unpack and view the raw dataset.	Use **Tableau Cloud User Permissions** to encrypt connections and restrict access to sensitive commercial manufacturer data.

FutureRoadmap:

Phase	Planned Feature / Enhancement	TargetTimeline	Expected Impact
Phase 2	Automated Pipelines & Cloud Hosting	1 Month	Transition data to a cloud warehouse with automated weekly schedules, freeing up the developer from manual file updates.
Phase 3	advanced Predictive Analysis	3 Month	Integrate Tableau with **Python/R (TabPy)** to build battery life decay projections and forecasting charts for market sales.
Phase 4	Real-Time Infrastructure Mapping	6 Month	Implement live API data feeds connecting regional charging network station metrics to map out real-time infrastructure growth gaps.